CALCULUS · UNIT 5: ANTIDERIVATIVES AND DEFINITE INTEGRALS

Solar Energy Audit

Use Riemann sums, antiderivatives, and definite integrals to measure a school's solar energy.

Honors

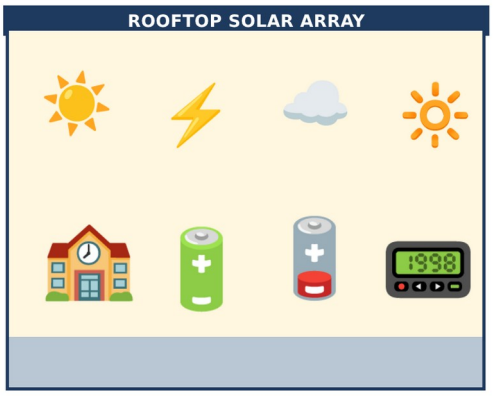
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Your school installed solar panels on the roof. A meter records the power output (in kilowatts) every two hours, but the principal wants to know the total energy (in kilowatt-hours) produced each day.

Energy is the accumulation of power over time. You will estimate and calculate it with Riemann sums and integrals.



Units

Power in kW

Energy in kWh

Goal

Cover 600 kWh

per school day

Learning Goals

- I can analyze Riemann sums and their error.
- I can evaluate integrals using antiderivatives and u-substitution.
- I can build and compare integral models.

Project Timeline

When	What you will do
Day 1	Error Analysis
Day 2	Better Models
Day 3	Upgrade Analysis
Final	Energy Audit Report + reflection

Part 1: Error Analysis

Day 1 · Use the Power Meter Readings and $P(t) = -2t^2 + 24t$.

Power Meter Readings (kW)

6 a.m.	8 a.m.	10 a.m.	12 p.m.	2 p.m.	4 p.m.	6 p.m.
0	24	58	72	60	30	4

1. Compute the trapezoidal sum with the data and with the model values at t = 0, 2, …, 12.

2. Find the exact integral of the model and the error of the model's trapezoidal sum. Why is it an underestimate?

3. Write the limit of a right Riemann sum that equals $\int_0^{12} (-2t^2 + 24t)\,dt$.

4. Which data reading disagrees most with the model? What might explain it?

Part 2: Better Models

Day 2 · A bell-shaped model: Q(t) = 72 sin($\frac{\pi t}{12}$) kW.

5. Find $\int_0^{12} 72 \sin(\frac{\pi t}{12})\, dt$ using substitution.

6. Compare Q with the data at 8 a.m. and 4 p.m. Which model fits better?

7. Find ∫ t e^(−t²) dt using u-substitution and explain each step.

8. Find the average power over the day using Q.

Part 3: Upgrade Analysis

Day 3 · Goal: 600 kWh per day using model Q.

9. How many 4% panels are needed so the energy reaches 600 kWh?

10. Battery storage starts charging when power exceeds 40 kW. Find when Q(t) > 40 and the energy above 40 kW.

11. Write the audit report comparing the models, with error analysis and your recommendation.

Final Product: Energy Audit Report

Your final product must include:

- Trapezoid error analysis
- Limit-of-sums definition
- Sine model with u-substitution
- Upgrade and storage analysis
- Audit report

Reflection

R1. How does concavity tell you whether a trapezoidal sum over- or underestimates?

R2. Why choose one model over another when both are reasonable?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Riemann sums, antiderivatives, and definite integrals are computed correctly and used to judge the energy goal. Goes beyond the minimum with a creative, realistic choice.	Riemann sums, antiderivatives, and definite integrals are computed correctly and used to judge the energy goal.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Audit report includes data, sums, integrals, graphs, and a recommendation. Polished, organized, and visually engaging.	Audit report includes data, sums, integrals, graphs, and a recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.